

AI Model Comparison for Web-Crawler Project

1. Overview

Your web-crawler needs two main AI functions:

1. Embeddings
Convert each scraped webpage into numerical vectors for search, clustering, and indexing.
2. Summaries / Content Understanding
Short, clear summaries of each page so they can be classified, ranked, or quickly reviewed.

We compared three providers:

- OpenAI API — <https://platform.openai.com/docs/pricing>
- Google Gemini API — <https://ai.google.dev/gemini-api/docs/pricing>
- DeepSeek API — https://api-docs.deepseek.com/quick_start/pricing/

Below are the most compatible models, their costs, and which one is best for your use case.

2. Most Compatible Models (Simple Table)

A. Model Comparison — Best Models for Web Crawlers

Generation / Summaries

Provider	Model	Input Cost (per 1M tokens)	Output Cost (per 1M tokens)	Why It's Good
Google Gemini	gemini-2.5-flash-lite	~\$0.10	~\$0.40	Very cheap, fast, reliable for bulk summaries.
OpenAI	gpt-5-mini	~\$0.25	~\$2.00	Good balance of quality and price.
OpenAI	gpt-4.1-mini	~\$0.40	~\$1.60	Higher reasoning ability than 5-mini.

Provider	Model	Input Cost (per 1M tokens)	Output Cost (per 1M tokens)	Why It's Good
DeepSeek	deepseek-chat (cache miss)	~\$0.28	~\$0.42	Low cost even without caching.
DeepSeek	deepseek-chat (cache hit)	~\$0.028	~\$0.42	Extremely cheap with caching enabled.

Links:

OpenAI — <https://platform.openai.com/docs/pricing>

Gemini — <https://ai.google.dev/gemini-api/docs/pricing>

DeepSeek — https://api-docs.deepseek.com/quick_start/pricing/

Embeddings (for semantic search and indexing)

Provider	Model	Cost per 1M tokens	Notes
OpenAI	text-embedding-3-small	~\$0.02	Best price-to-performance for large crawls.
OpenAI	text-embedding-3-large	~\$0.13	Higher accuracy for search/classification.
Google Gemini	gemini-embedding-001	~0.15 (approx.)	Good quality, slightly higher cost.
DeepSeek	No separate embedding model	Uses chat input pricing	Works, but not optimized for embeddings.

Feature Comparison

Feature	OpenAI API	Google Gemini API	DeepSeek API
Embeddings support	Yes (text embeddings)	Yes (Gemini embedding models)	Yes (via chat model)
Text summarization / generation	Yes	Yes (Gemini models do responses)	Yes (chat/reasoner models)
Multimodal support (images, video)	Limited / model-dependent	Yes (Gemini supports multimodal)	Usually text only
Easy to integrate with existing OpenAI SDKs	Yes	Some differences (Google client SDKs)	Yes (API is OpenAI-compatible)
Best for large context (long documents)	Yes (long context models like GPT-4.1)	Yes, Google supports large context & function calls	Supports up to ~128K context via model versions
Lowest cost per token generally	Varies by model — can be expensive at scale (depends on usage)	Varies by model; Google promotional credits often available	Often cheaper or competitive (aggressive pricing)
Enterprise plans / SLA	Yes (enterprise SLA available)	Yes (Google Cloud enterprise SLAs)	Not as mature publicly documented

3. Best Fit Recommendations

Overall Best Value:

Gemini's gemini-2.5-flash-lite + OpenAI text-embedding-3-small → lowest operating cost while keeping accuracy high.

Best Quality Summaries:

OpenAI gpt-5-mini → higher reasoning and cleaner summaries than Gemini Flash-Lite.

Lowest Cost at Scale:

DeepSeek-chat (CACHE HIT pricing) → cheapest option when pages contain similar or repeated content.

Best Embeddings:

OpenAI text-embedding-3-large → highest accuracy for semantic search.

4. Simple Cost Example

Assumption:

- Average page = 2 KB $\approx$ 500 tokens
- Summary = $\sim$ 200 output tokens

Example for 100,000 pages/month

Provider	Summary Model	Embedding Model	Est. Monthly Cost
Gemini	2.5-flash-lite	gemini-embedding-001	Low ($\sim$ \$150–\$300)
OpenAI	gpt-5-mini	text-embedding-3-small	Medium ($\sim$ \$300–\$500)
DeepSeek	deepseek-chat	deepseek-chat (input cost)	Very Low if cache hit ($<$ \$100)

(These are approximations — actual usage will vary.)

5. Which Should We Choose?

Recommended Setup

Gemini 2.5-Flash-Lite (summaries) + OpenAI text-embedding-3-small (embeddings)

- ✓ Lowest combined cost
- ✓ Fastest output
- ✓ Easy integration
- ✓ Good accuracy for search + categorization

If we need *higher quality summaries*, select OpenAI gpt-5-mini instead.

If we want *the absolute lowest cost* and pages are repetitive, enable DeepSeek caching.

Conclusion:

For a web-crawler pipeline that needs to process large numbers of pages every month, generate summaries, and produce embeddings for search and classification, **Google Gemini's gemini-2.5-flash-lite** stands out as the most compatible and cost-efficient model among the three providers evaluated. The core reason is that Flash Lite is specifically optimized for high-volume, text-heavy workloads. Web crawlers deal with many small documents, and this model handles those short inputs quickly, consistently, and at a much lower cost per token than most comparable models from OpenAI or DeepSeek.

Flash Lite provides strong, reliable summarization and can interpret messy, real-world webpage text without requiring a heavier, more expensive model. Its token pricing is significantly lower than equivalent OpenAI models, and its latency profile makes it well-suited for pipelines that need to keep up with continuous crawling. The model also supports embeddings through the same API ecosystem, allowing the crawler to extract meaning, classify pages, or feed a semantic search system without switching between different services or SDKs. That reduces integration work, simplifies maintenance, and keeps infrastructure lightweight.

In terms of scalability, Gemini's infrastructure allows the system to handle bursts of traffic and process large batches of pages with minimal slowdown. This matters because crawling workloads are rarely stable — some days require processing thousands of pages within minutes, and Flash Lite can comfortably sustain those surges. If the project expands later, Gemini also offers multimodal capabilities for images, screenshots, or visual data extracted from pages. This means the system can evolve without changing providers or redesigning the pipeline.

Overall, when balancing cost, performance, ease of integration, and long-term flexibility, **Gemini 2.5 Flash Lite provides the best fit for the project's requirements**. It meets the technical needs of a modern crawler while keeping operating costs predictable and low, making it the most practical and future-ready choice among the available AI platforms.

References:

OpenAI

OpenAI Platform – Models & Pricing

<https://platform.openai.com/docs/models>

OpenAI Platform – Embeddings

<https://platform.openai.com/docs/guides/embeddings>

OpenAI Platform – Capabilities & Integrations

<https://platform.openai.com/docs>

Reuters – OpenAI Pricing Trends / Cost Competitiveness

<https://www.reuters.com/technology>

Google Gemini

Google AI for Developers – Gemini Models

<https://ai.google.dev>

Google AI for Developers – Multimodal & Text Capabilities

<https://ai.google.dev/gemini-api/docs>

Google AI for Developers – Embeddings

https://ai.google.dev/tutorials/python_quickstart

Aloa – Context Length Comparison Across Models

<https://aloe.co/blog>

Google Cloud – Enterprise SLAs

<https://cloud.google.com/terms/sla>

DeepSeek

DeepSeek API Docs – Models & Pricing

<https://api-docs.deepseek.com>

DeepSeek – Chat & Reasoning Model Capabilities

<https://api-docs.deepseek.com/api-reference/chat-completions>

DeepSeek – Integration (OpenAI-Compatible API)

https://api-docs.deepseek.com/quick_start

